Lab no.01

Ubuntu Server Installation

Submitted To: Engr.Waqas Saleem

Submitted By: Faiqa Shahzad

Reg No: 2024-BSE-26

Course Code: SE-665

Section: B

GitHub Username: FaiqaShahzad026

Task 1:

Create and Configure Your GitHub Account

Description:

A GitHub account was created and configured using my actual name and a professional username. The same GitHub username was selected for use as my Ubuntu username.

Screenshot:

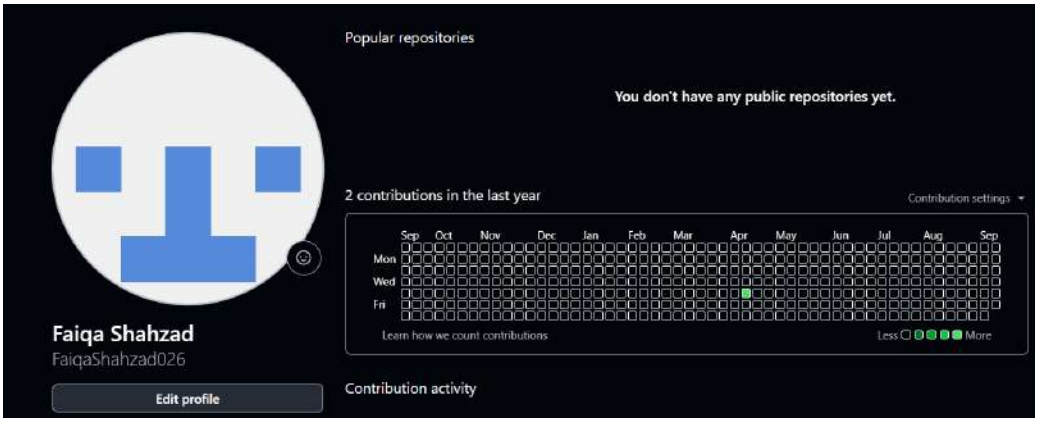


Figure 1.1 github_profile.png



Figure 1.2 github_actual_name.png

Task 2:

Register in the Student Marks Portal

Description:

I signed in to the Student Marks Portal using my GitHub account. My registration number, course code, and section were entered and the enrollment was submitted.

Screenshot:

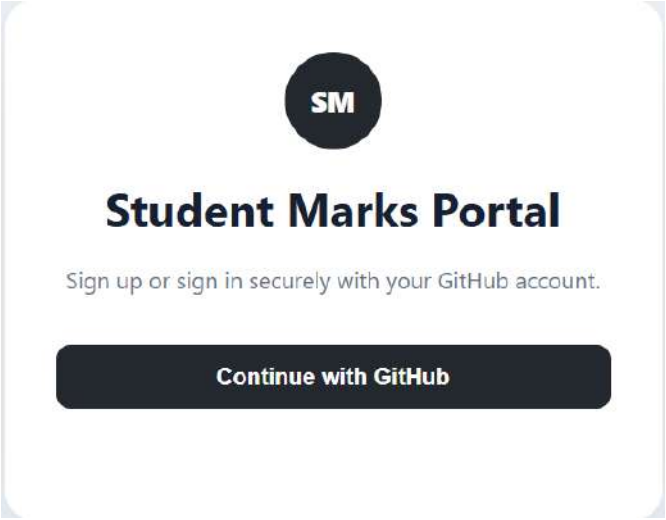


Figure 2.1 portal_login_page.png



Figure 2.2 portal_github_profile.png

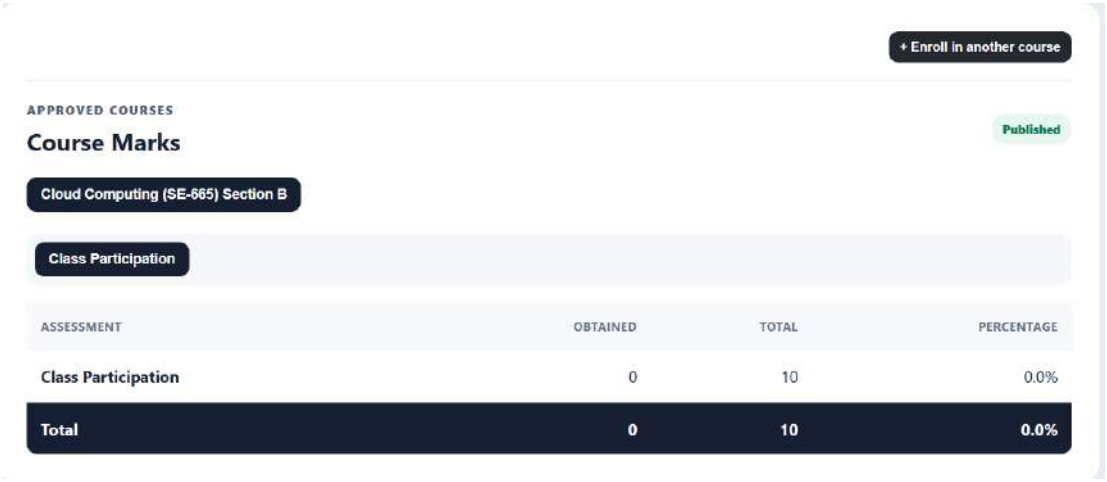


Figure 2.3 portal_enrollment_submitted.png

Task 3:

Download VMware Workstation Pro and Ubuntu Server

Description:

VMware Workstation Pro was installed and verified on the Windows computer. The Ubuntu Server LTS ISO was downloaded and sufficient storage space was checked.

Screenshot:

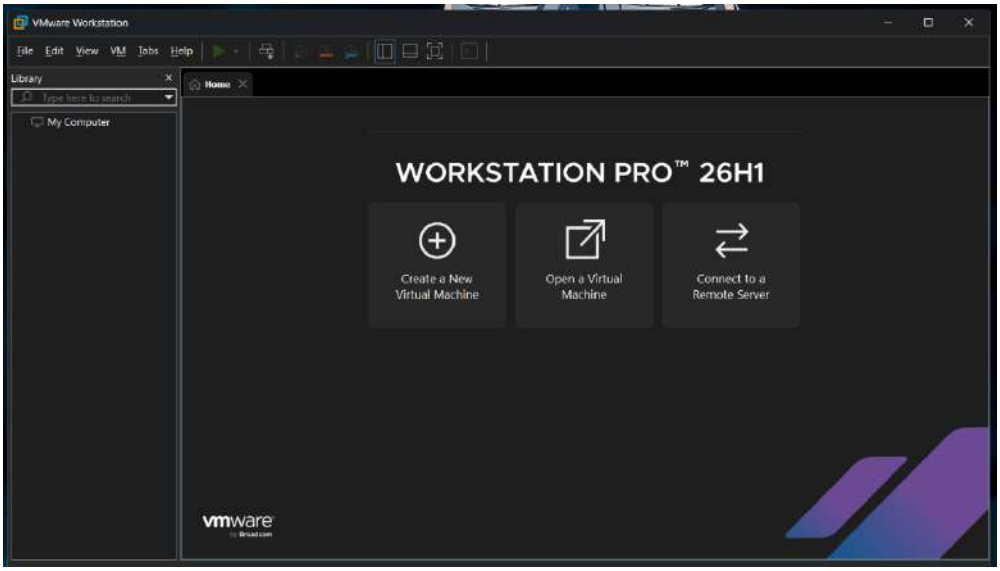


Figure 3.1 vmware_workstation.png

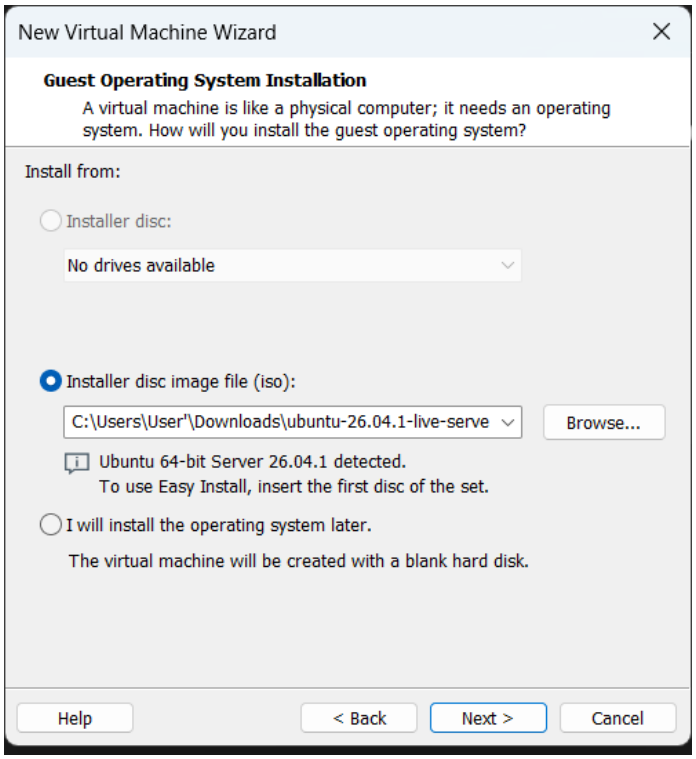


Figure 3.2 ubuntu_server_iso.png

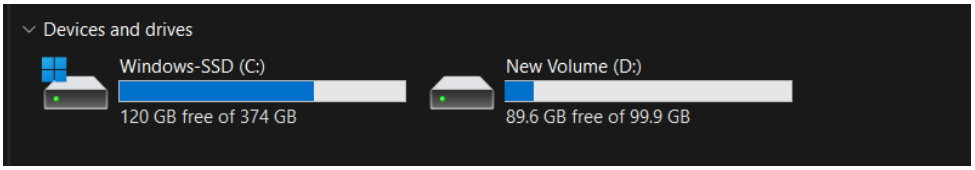


Figure 3.3 available_storage.png

Task 4:

Create the Ubuntu Server Virtual Machine

Description:

A new virtual machine was created in VMware Workstation Pro using the Ubuntu Server ISO. The VM was configured with the required virtual disk space and started successfully.

Screenshot:

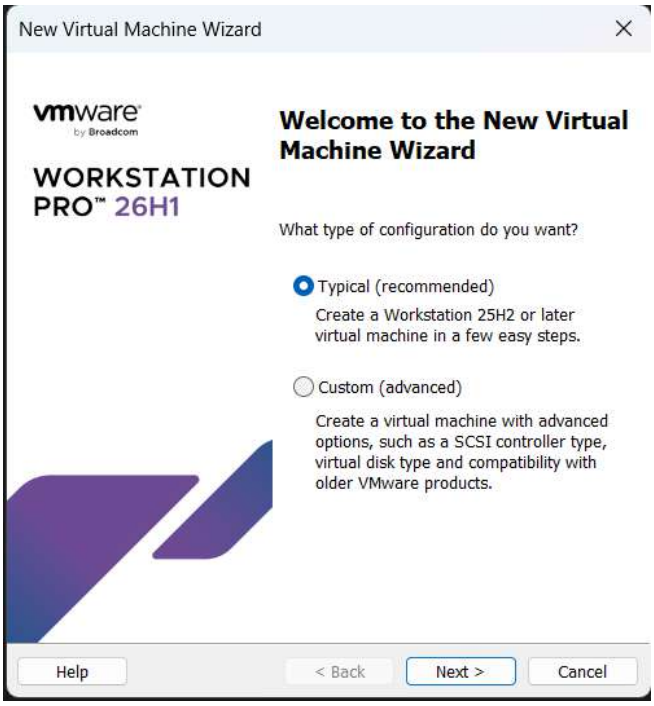


Figure 4.1 vm_creation_wizard.png

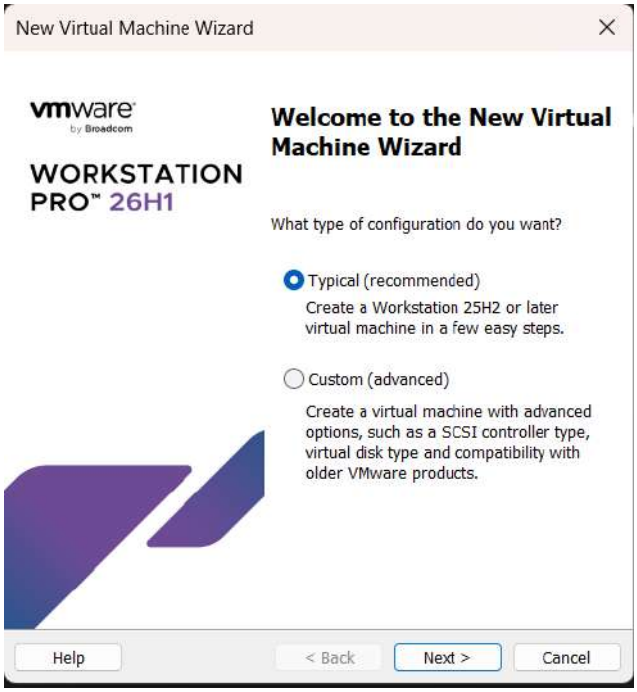


Figure 4.2 vm_typical_configuration.png

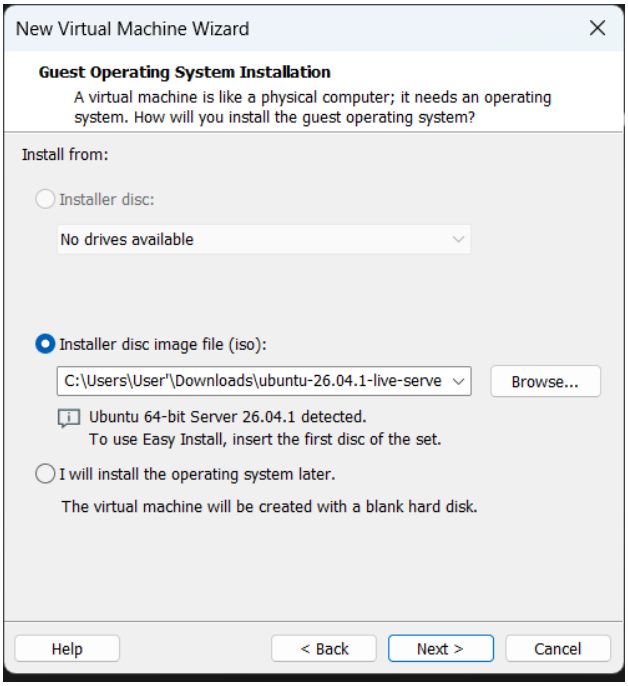


Figure 4.3 vm_ubuntu_iso_selected.png

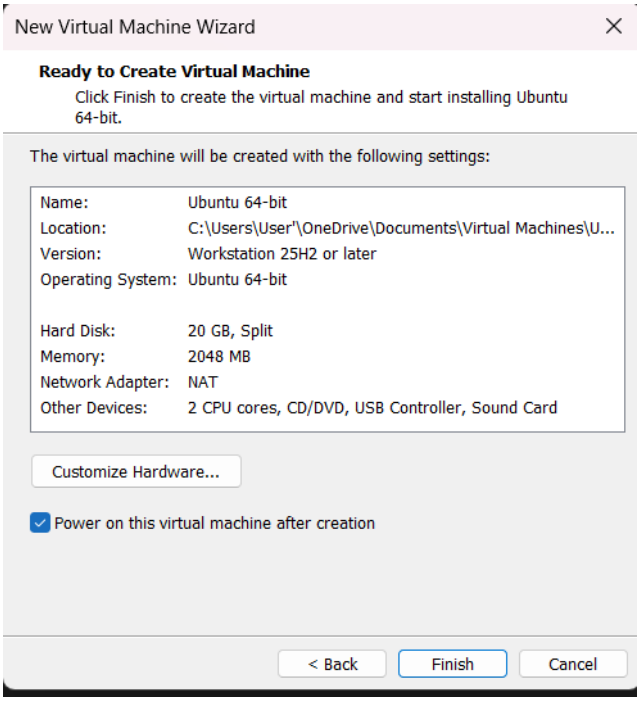


Figure 4.4 vm_configuration_summary.png

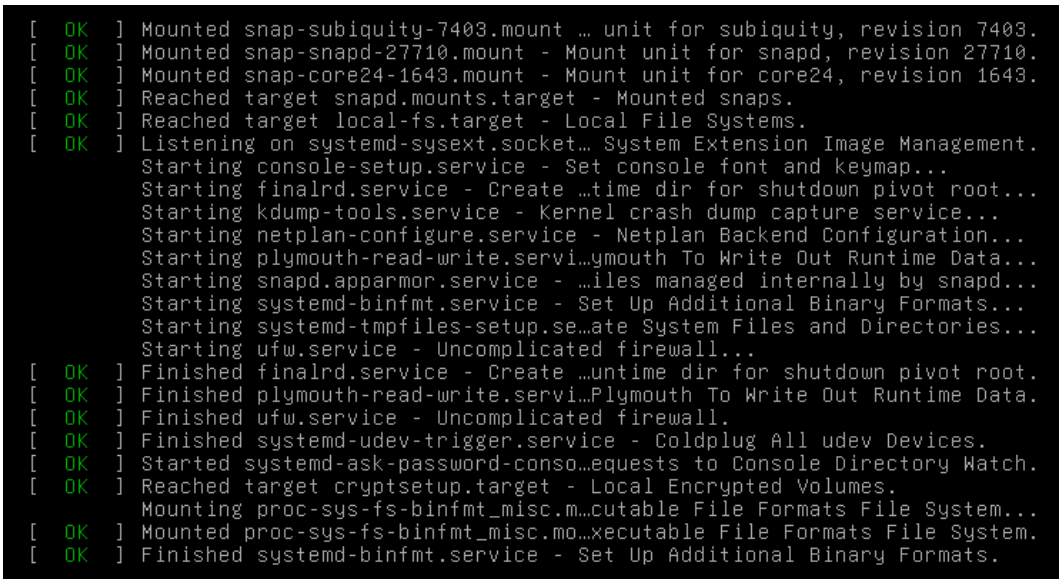


Figure 4.5 ubuntu_installer_boot.png

Task 5:

Choose the Installation Language

Description:

The required language was selected during the Ubuntu Server installation. The selected language was confirmed to continue with the installation.

Screenshot:

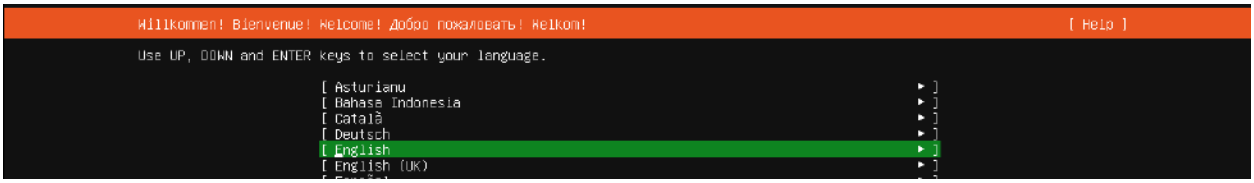


Figure 5.1 ubuntu_language.png

Task 6:

Configure the Keyboard Layout

Description:

The keyboard layout matching the computer keyboard was selected. The appropriate keyboard variant was also selected before continuing.

Screenshot:



Figure 6.1 ubuntu_keyboard_layout.png

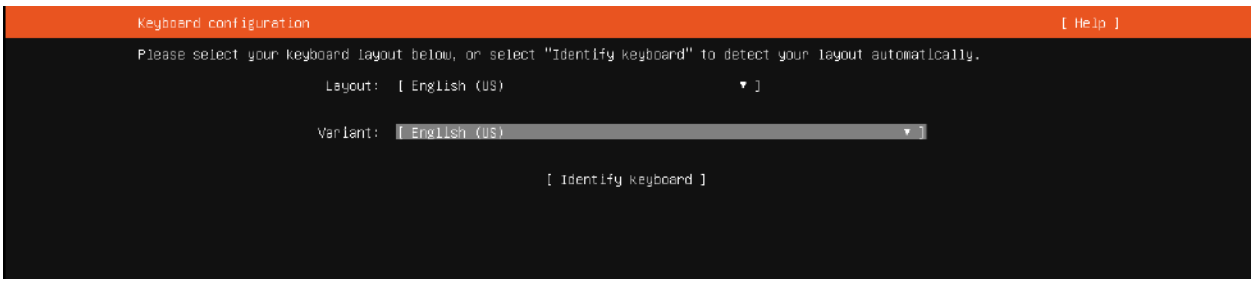


Figure 6.2 ubuntu_keyboard_variant.png

Task 7:

Choose the Ubuntu Server Installation Type

Description:

The normal Ubuntu Server installation option was selected. The MAAS installation option was not selected because it was not required for this lab.

Screenshot:

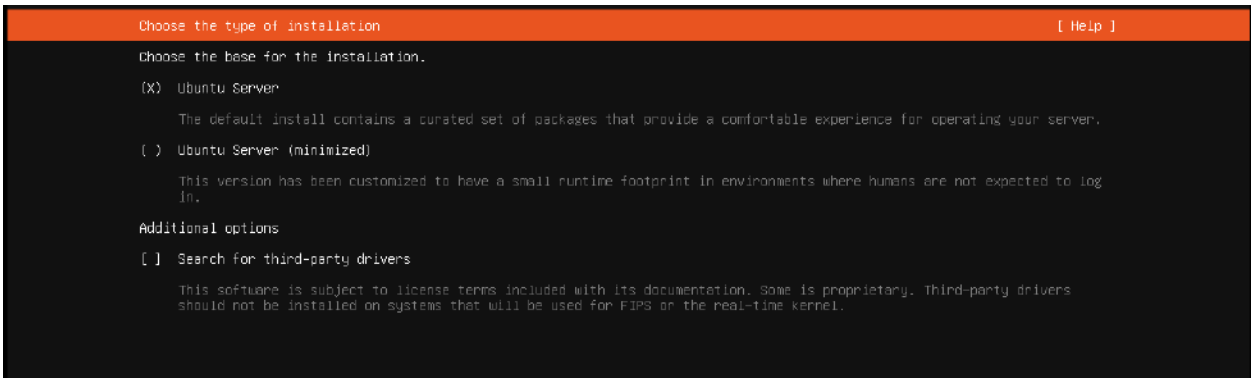


Figure 7.1 ubuntu_installation_type.png

Task 8:

Configure the Network

Description:

The network interface was detected and configured automatically using DHCP. The Ubuntu Server received an IPv4 address for network communication.

Screenshot:

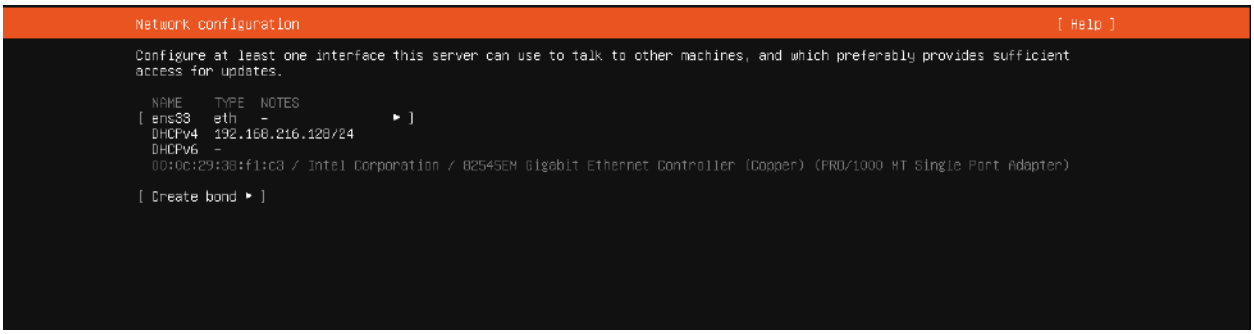


Figure 8.1 ubuntu_network_interface.png

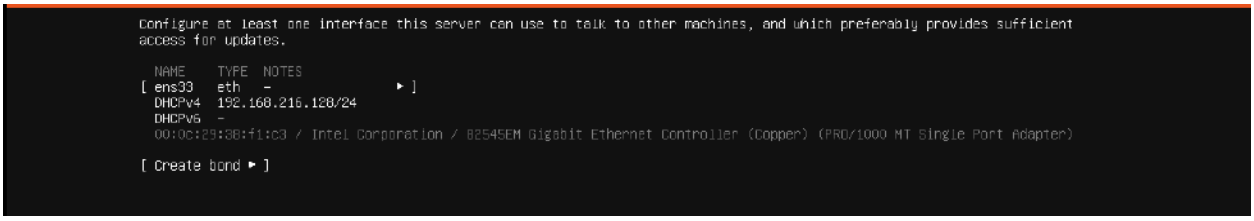


Figure 8.2 ubuntu_installer_network.png

Task 9:

Configure Storage

Description:

The guided storage option was selected to use the VMware virtual disk. The proposed storage configuration was reviewed before continuing.

Screenshot:

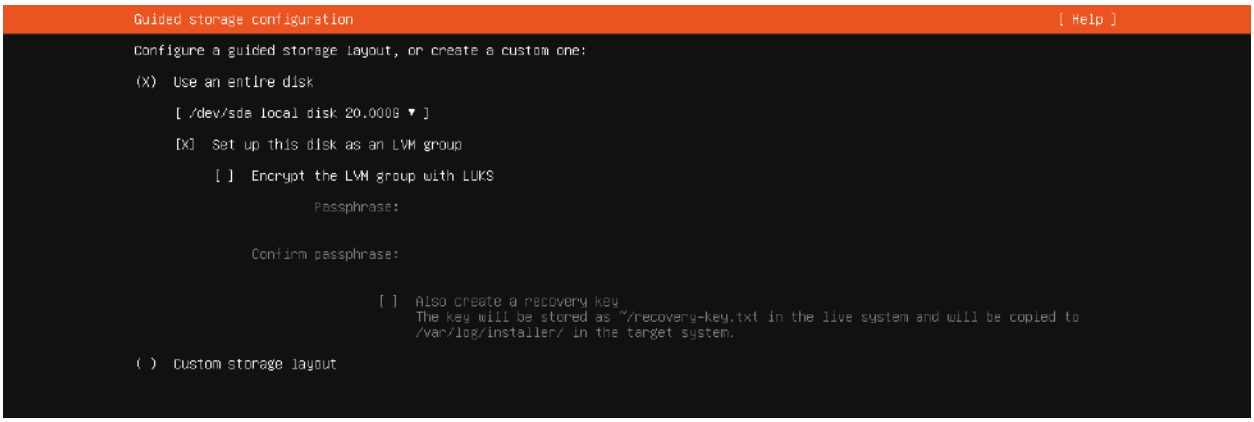


Figure 9.1 ubuntu_guided_storage.png

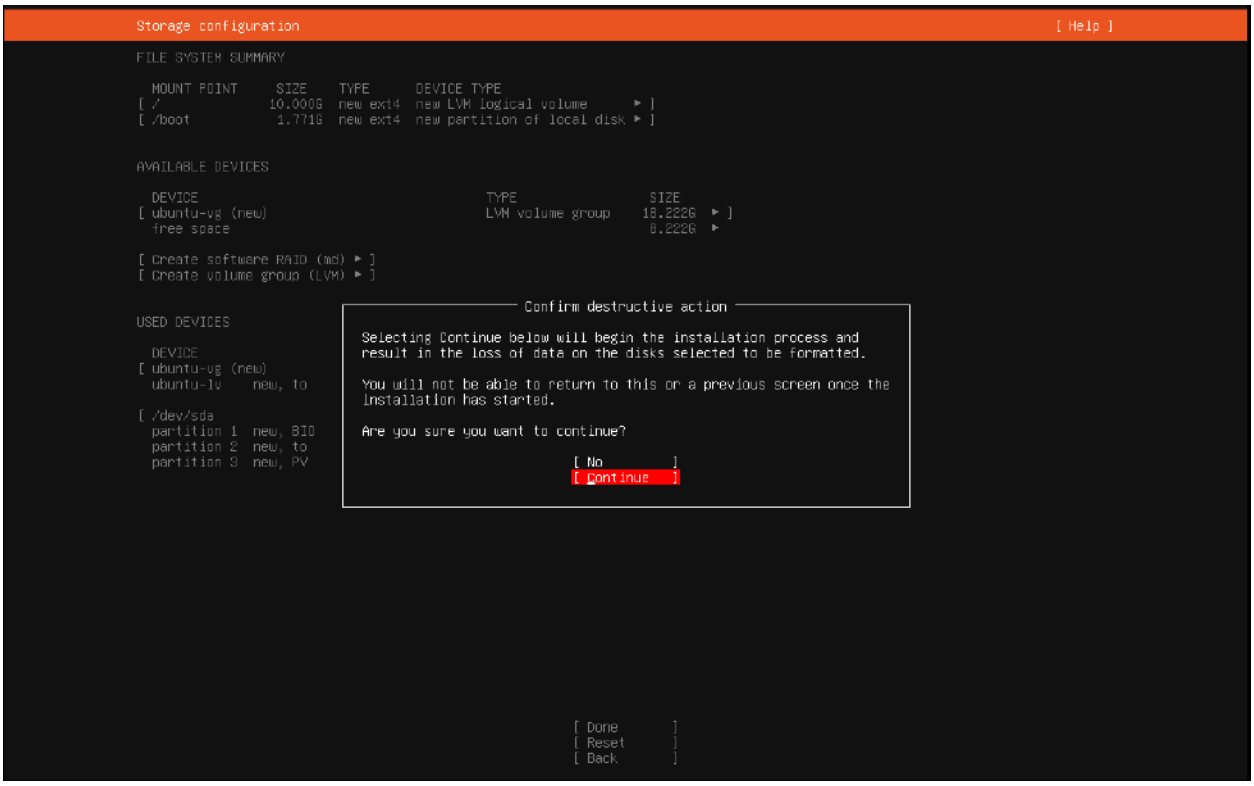


Figure 9.2 ubuntu_storage_configuration.png

Task 10:

Select the Installation Disk

Description:

The virtual disk created for the Ubuntu Server VM was selected as the installation disk. The disk name and size were checked to ensure the correct virtual disk was being used.

Screenshot:

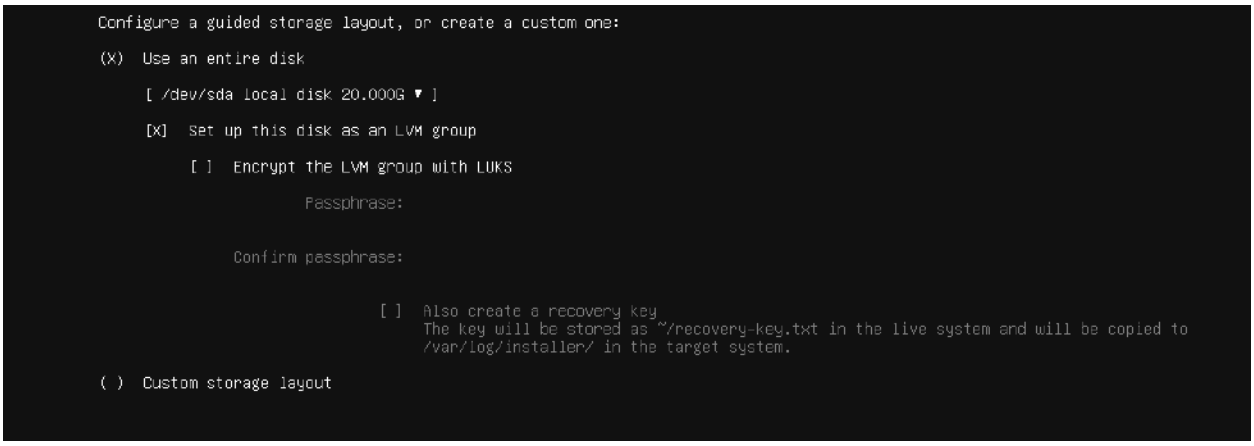


Figure 10.1 ubuntu_selected_disk.png

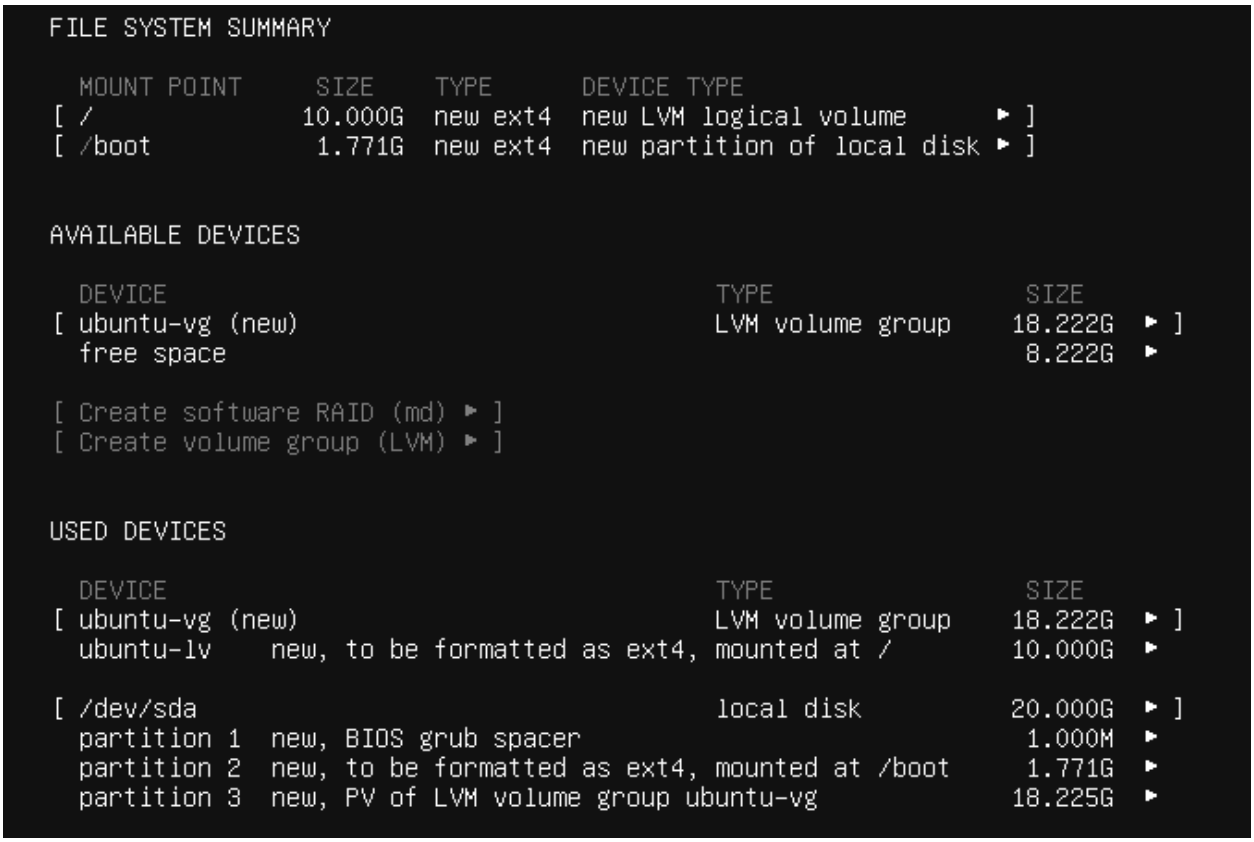


Figure 10.2 ubuntu_disk_details.png

Task 11:

Review the Partitions

Description:

The partition layout proposed by the Ubuntu installer was reviewed carefully. The partitions were verified to belong to the VMware virtual disk before proceeding.

Screenshot:

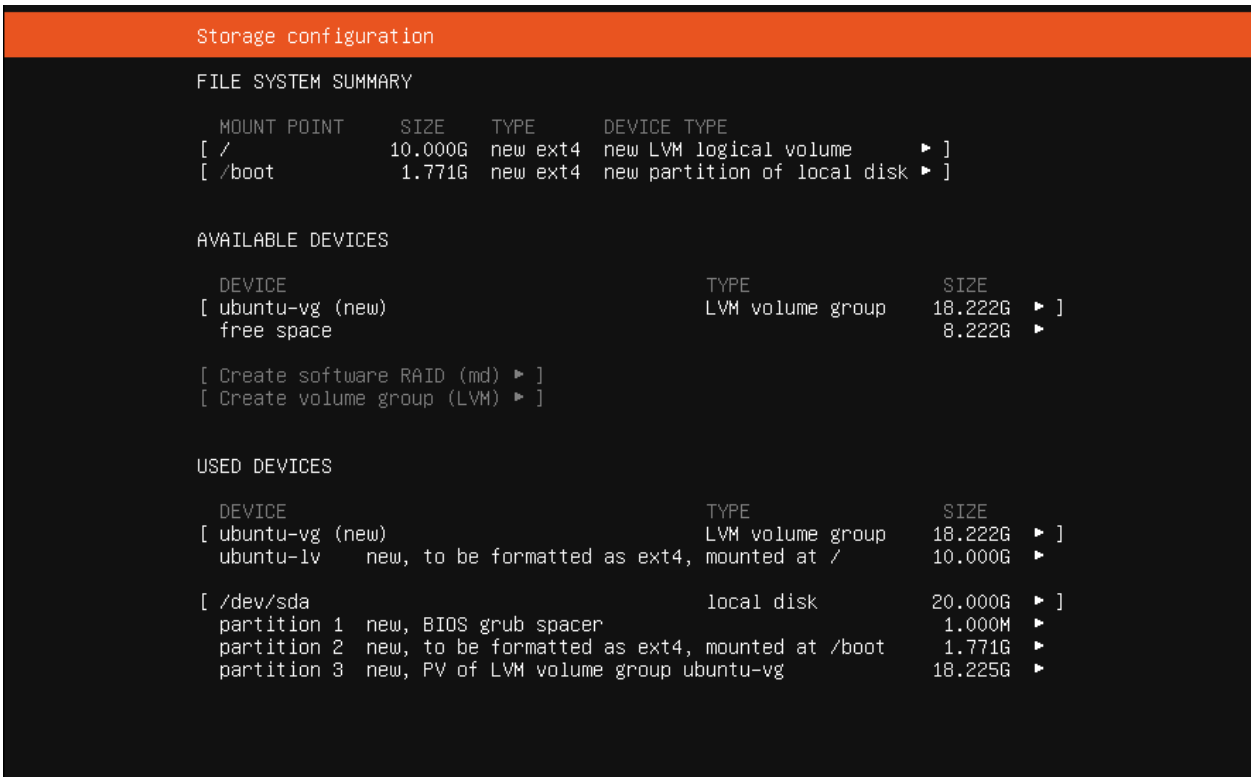


Figure 11.1 ubuntu_partition_layout.png

Task 12:

Confirm the Storage Changes

Description:

The storage warning was reviewed before applying the disk changes. The changes were confirmed after verifying that the correct VMware virtual disk was selected.

Screenshot:

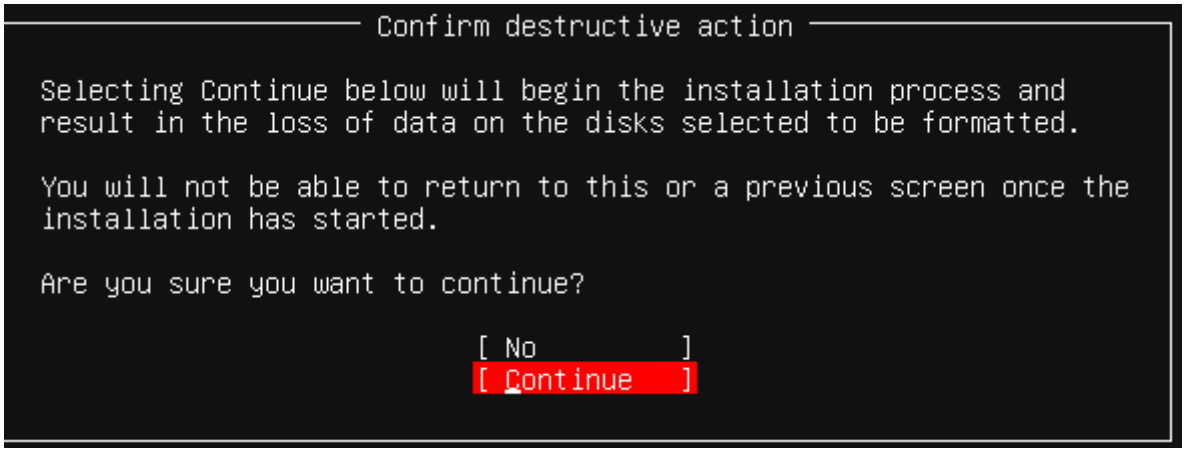


Figure 12.1 ubuntu_storage_warning.png

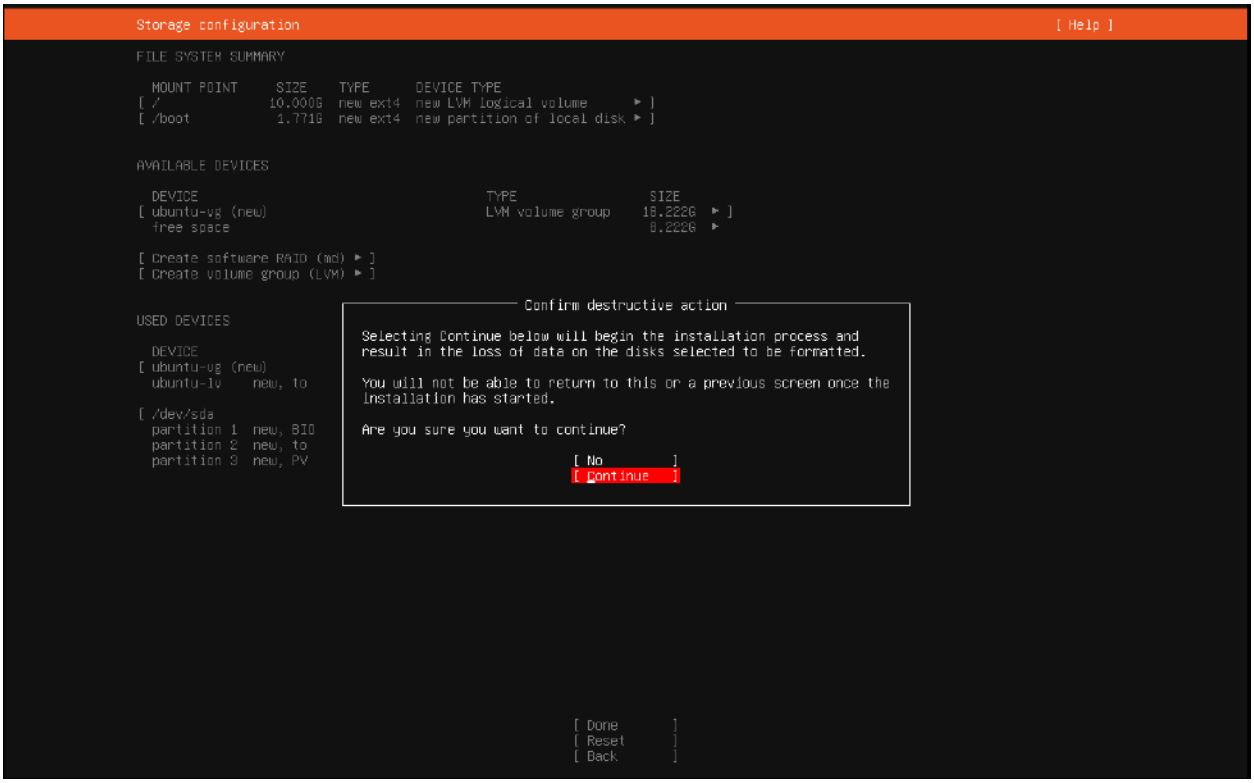


Figure 12.2 ubuntu_storage_confirmed.png

Task 13:

Configure the Ubuntu Profile

Description:

The Ubuntu profile was configured using my actual name, the server name ubuntu, and my GitHub username. A secure password was also created and confirmed.

Screenshot:

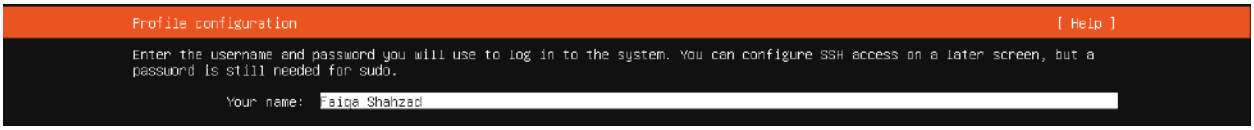


Figure 13.1 ubuntu_actual_name.png



Figure 13.2 ubuntu_server_name.png



Figure 13.3 ubuntu_username.png

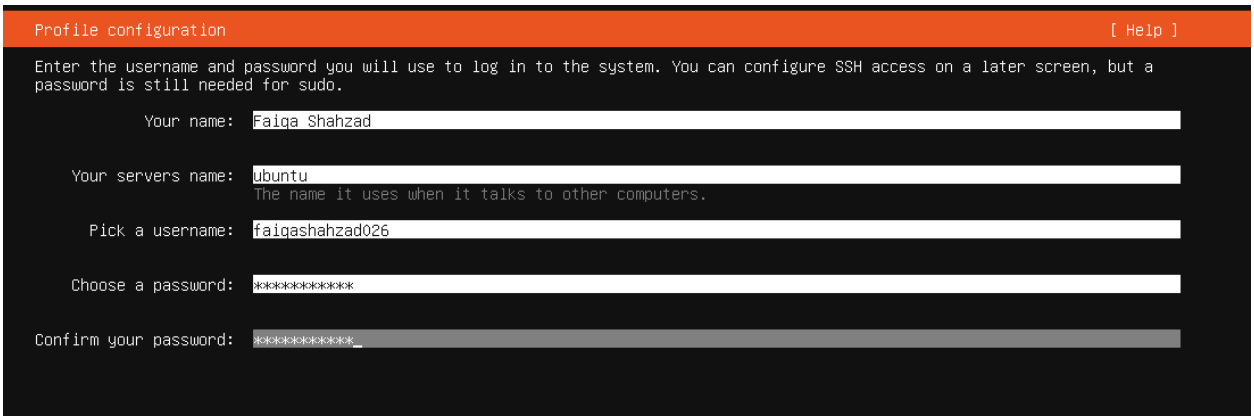


Figure 13.4 ubuntu_profile_setup.png

Task 14:

Complete the Software Installation

Description:

Ubuntu Server installed the required software and system components. The installation was allowed to complete without shutting down the virtual machine.

Screenshot:



Figure 14.1 ubuntu_installation_progress.png

Task 15:

Reboot and verify the Ubuntu Identity

Description:

After installation, the Ubuntu Server was rebooted and successfully logged in. The terminal prompt was verified to show my GitHub username with the server name ubuntu.

Screenshot:

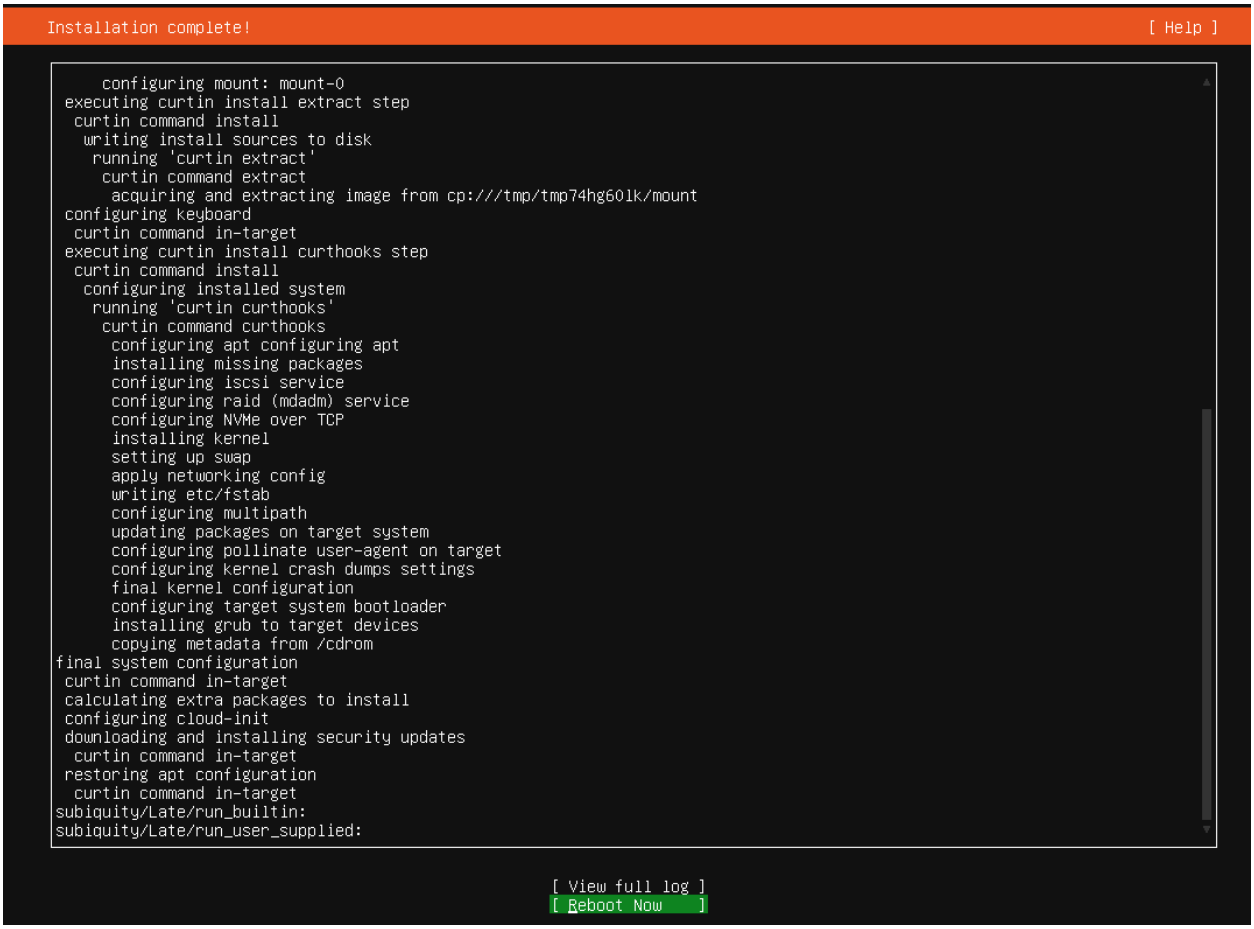


Figure 15.1 ubuntu_installation_complete.png

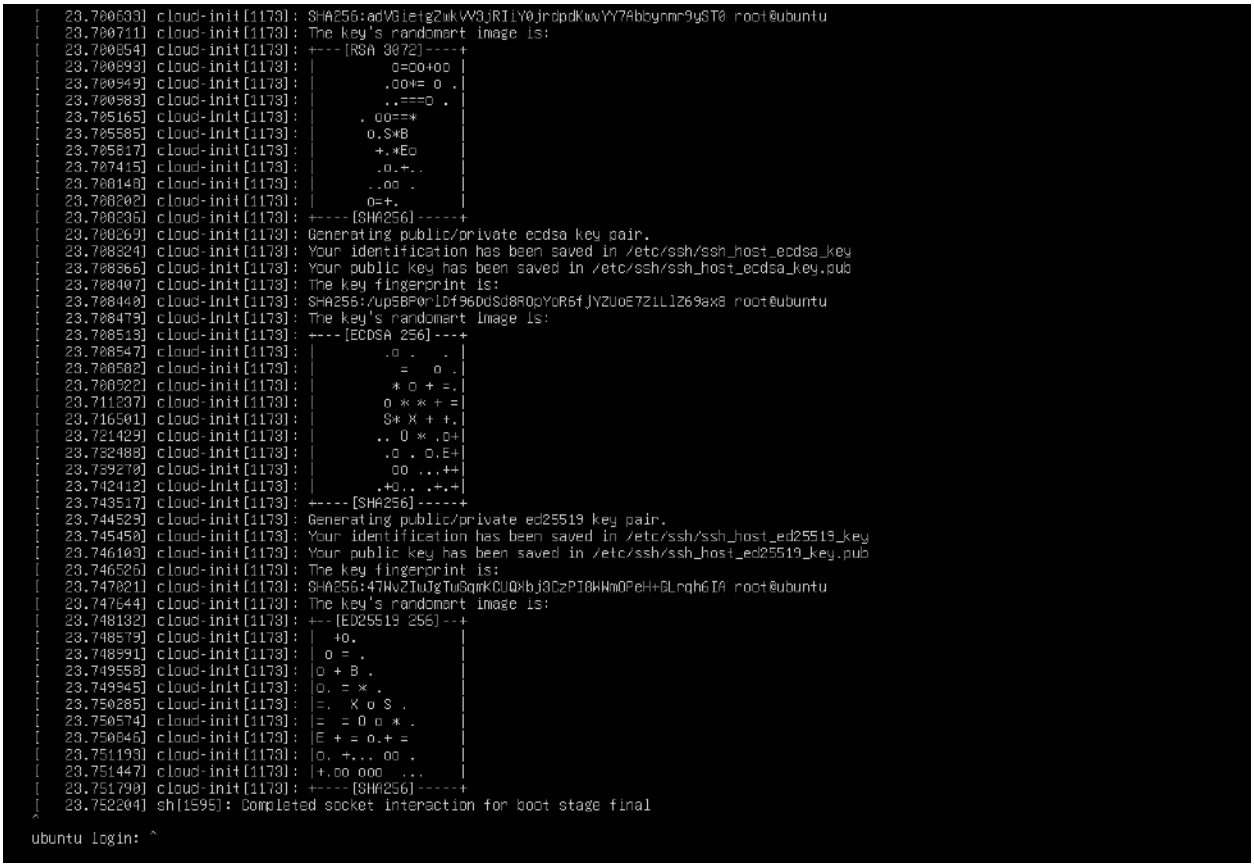


Figure 15.2 ubuntu_login_screen.png



Figure 15.3 ubuntu_terminal_login.png

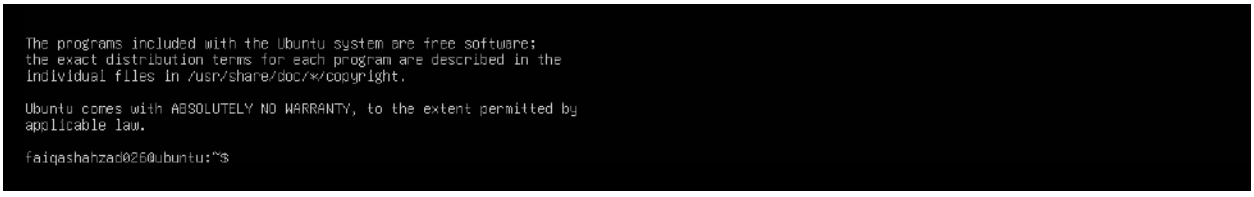


Figure 15.4 ubuntu_identity_verified.png

Task 16:

Find the Ubuntu Server IP Address

Description:

The ***ip addr*** command was used to identify the network configuration of the Ubuntu Server. The active network interface IPv4 address was identified for the SSH connection.

Screenshot:

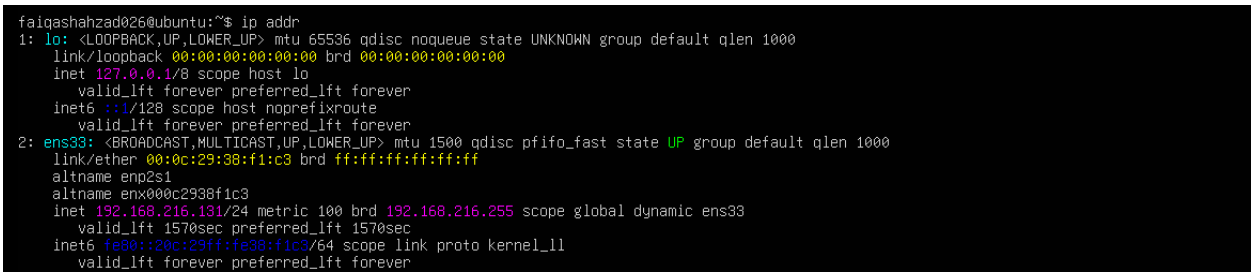


Figure 16.1 ubuntu_ip_addr_command.png

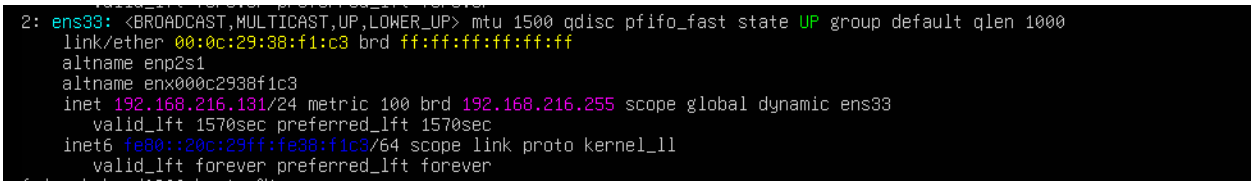


Figure 16.2 ubuntu_ip_address.png

Task 17:

Connect to Ubuntu Server from Windows

Description:

An SSH connection was established from Windows to the Ubuntu Server using its username and IP address. The successful remote login was verified by checking the ***faiqashahzad026@ubuntu*** terminal prompt.

Screenshot:

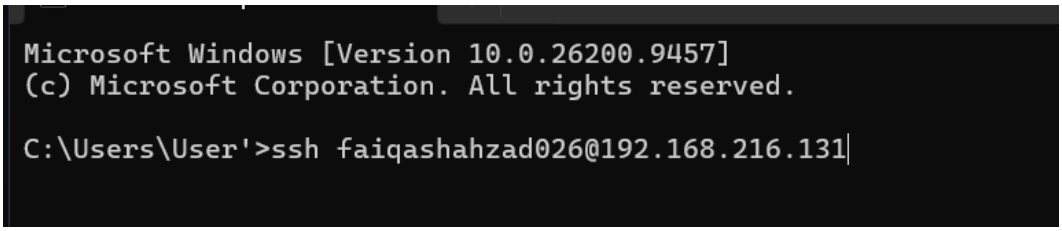


Figure 17.1 windows_ssh_command.png

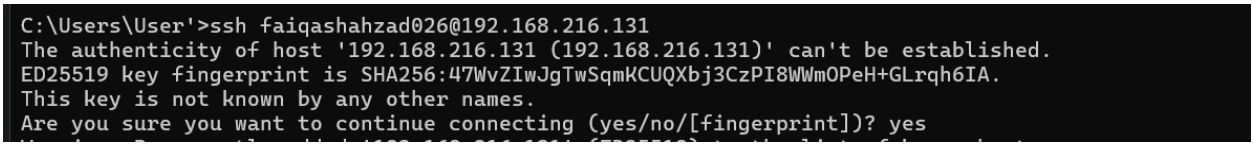


Figure 17.2 windows_ssh_fingerprint.png

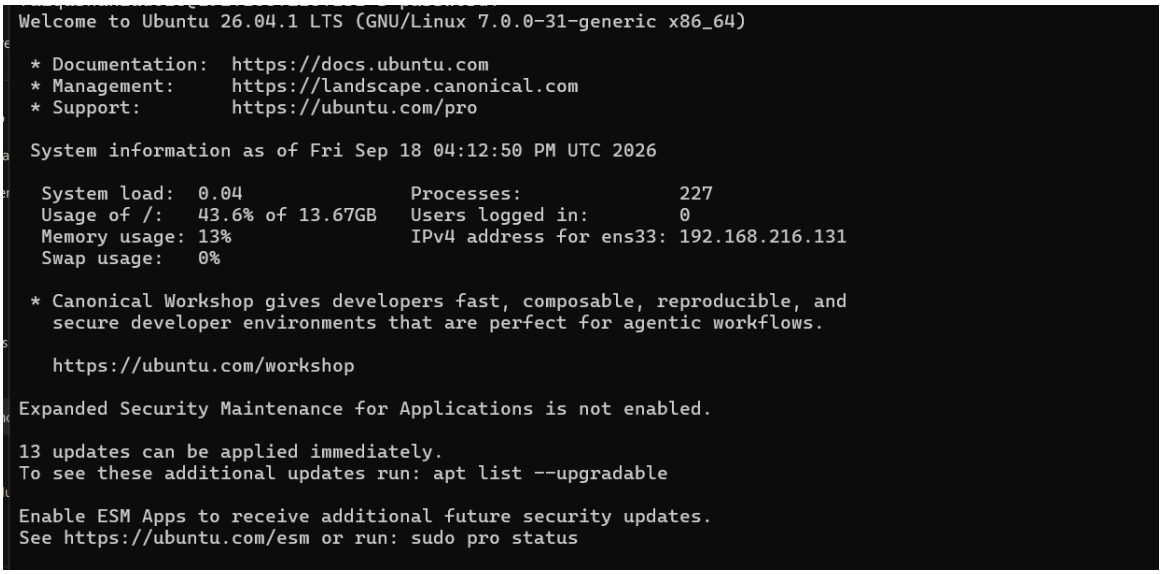


Figure 17.3 windows_ssh_login.png

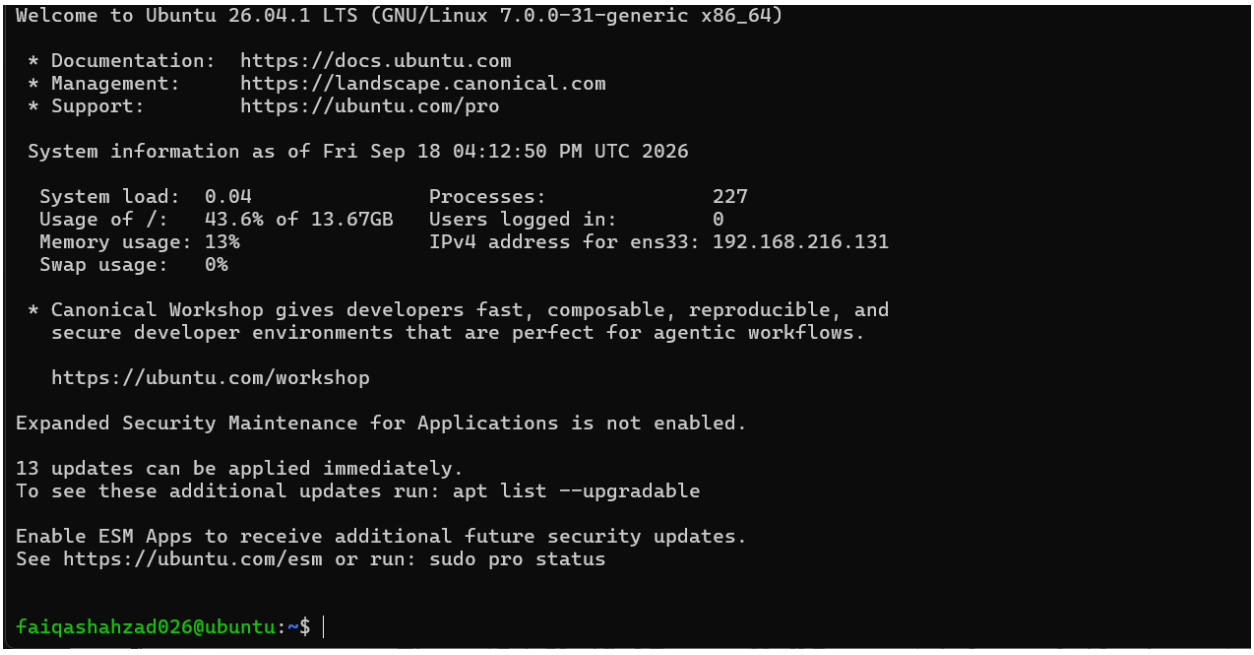


Figure 17.4 windows_ssh_identity.png

Task 18:

Prepare Lab1 Solution PDF

Lab no.01

Ubuntu Server Installation

Submitted To: Engr.Waqas Saleem

Submitted By: Faiqa Shahzad

Reg No: 2024-BSE-26

Course Code: SE-665

Section: B

GitHub Username: FaiqaShahzad026